

Linux Fundamentals – File Sharing

1. What is a Server?

- A server is a computer that provides services and resources to other computers connected to a network.
 - It is generally more powerful than the client systems on the network.
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2. Types of Servers

Different servers perform different tasks:

- **Print Server** – Manages printing services.
 - **Application Server** – Hosts and runs applications.
 - **Database Server** – Stores and manages databases.
 - **Email Server** – Handles email communication.
 - **Web Server** – Hosts websites on the internet.
 - **Authentication Server** – Verifies user identity.
 - **File Server** – Stores and shares files across a network.
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3. Print Server

- Manages print jobs in a network.
 - Routes print requests to the appropriate printer.
 - Prioritizes print jobs based on the order of requests.
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4. File Server

- Stores files in a central location.
 - Allows users to access and share files over a network.
 - Eliminates the need to physically transfer files.
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5. Authentication Server

- Verifies user credentials before granting access.
- Ensures secure access to networks, systems, applications, or services.
- Acts as a security layer for authentication.

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6. Database Server

- Stores and manages organizational databases.
 - Handles database storage, queries, and data retrieval.
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7. Email Server

- Manages incoming, outgoing, and internal emails.
 - Remains connected to the internet for email communication.
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8. Web Server

- Hosts websites and web applications.
 - Makes websites accessible through the internet.
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9. Python Simple HTTP Server

- A lightweight HTTP server module in Python.
- Converts any directory into a simple web server.
- Supports only **GET** and **HEAD** HTTP methods.
- Useful for sharing files over a network.
- In Python 3, it is available through the `http.server` module.

Commands

```
sudo apt-get install python3  
python3 -m http.server 9000
```

10. Apache Web Server

- Apache is a free and open-source web server.
- Maintained by the Apache Software Foundation.
- Used to deploy websites on the internet.
- One of the oldest and most reliable web server software.

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Installation

```
sudo apt install apache2
```

Service Management

```
sudo systemctl enable apache2
sudo systemctl disable apache2
sudo systemctl start apache2
sudo systemctl stop apache2
```

Testing the Server

- Open **http://localhost** in the browser.
- The default Apache page confirms successful installation.

Website Directory

- Default location:

```
/var/www/html
```

Replace the default `index.html` with your own website files.

11. Samba

- Samba is an open-source software suite.
- Provides file and printer sharing across different operating systems.
- Allows Linux, Windows, and macOS systems to share files over a network.

What You'll Learn

- Setting up a Samba file server.
 - Sharing files across a local network.
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12. Installing Samba

```
sudo apt update
sudo apt install samba
```

Verify Installation

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whereis samba

13. Managing Samba Service

Check service status:

```
sudo systemctl status smbd
```

Start the service:

```
sudo systemctl start smbd
```

Enable the service at boot:

```
sudo systemctl enable smbd
```

14. Setting up Samba

Create a shared directory:

```
mkdir /home/<username>/sambashare/
```

Open the configuration file:

```
sudo nano /etc/samba/smb.conf
```

15. Samba Configuration

Add the following configuration:

```
[sambashare]
comment = Samba on Ubuntu
path = /home/username/sambashare
read only = no
browsable = yes
```

After editing:

- Press **Ctrl + O** to save.
 - Press **Ctrl + X** to exit Nano.
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16. Creating a Samba User

```
sudo smbpasswd -a username
```

17. Accessing Samba

- **Ubuntu:** File Manager → Connect to Server.
 - **macOS:** Finder → Go → Connect to Server.
 - **Windows:** Open File Explorer and enter the shared network path.
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18. NFS (Network File System)

- NFS is a protocol for sharing directories and files between Linux systems.
 - The shared directory is created on the NFS server.
 - This shared directory is also called the **export directory**.
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19. Installing NFS Server

```
sudo apt-get update  
sudo apt install nfs-kernel-server
```

20. Creating the Shared Directory

Create the export directory:

```
sudo mkdir /mnt/nfs_share
```

Set permissions:

```
sudo chown nobody:nogroup /mnt/nfs_share  
sudo chmod 777 /mnt/nfs_share
```

These permissions allow all client systems to access the shared directory.

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21. Define Client Access

Client access is configured in:

```
/etc/exports
```

You can allow:

- A single client
 - Multiple clients
 - An entire subnet
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22. Export Configuration

Example:

```
/mnt/nfs_share client_IP_1(rw,sync,no_subtree_check)
```

Options

- **rw** – Read and write permission.
 - **sync** – Writes changes to disk before applying them.
 - **no_subtree_check** – Disables subtree checking.
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23. Export the Shared Directory

```
sudo exportfs -a  
sudo systemctl restart nfs-kernel-server
```

- Makes the shared directory available.
 - Restarts the NFS server.
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24. NFS Client Setup

Install the required package:

```
sudo apt update  
sudo apt install nfs-common
```

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25. Create Mount Point

```
sudo mkdir -p /mnt/nfs_clientshare
```

26. Mount the Shared Directory

```
sudo mount 10.0.2.15:/mnt/nfs_share /mnt/nfs_clientshare
```

This allows the client machine to access the shared directory from the NFS server.

Key Takeaways

- Understood the purpose of different server types.
- Learned to create a simple Python HTTP server.
- Installed and configured Apache Web Server.
- Explored Samba for file sharing across different operating systems.
- Learned how NFS enables directory sharing between Linux systems.
- Practiced the commands required to configure and manage file-sharing services.